

Non-Line-of-Sight Transient Rendering

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ABSTRACT

Transient imaging methods often analyze time-resolved light transport for applications such as range imaging, reflectance estimation and, especially, non-line-of-sight (NLOS) imaging, which targets the reconstruction of hidden geometry using measurements of indirect diffuse reflections emitted by a laser. Transient rendering is a key tool for developing such new applications. In this work, we introduce a set of simple, yet effective subpath sampling techniques targeting transient light transport simulation in occluded scenes. We analyze the usual capture setups of NLOS scenes, where the light and camera indirectly aim at hidden geometry through a secondary path space. We leverage that configuration to reduce the integration path space. We implement our techniques in our modified version of Mitsuba 2, adapted for transient light transport, allowing us to support parallelization, polarization, and differentiable rendering.

CCS CONCEPTS

• Computing methodologies → Ray tracing.

KEYWORDS

light transport, transient rendering, non-line-of-sight imaging

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1 INTRODUCTION

Transient imaging aims to capture and analyze how light propagates through a scene with ultra-high temporal resolution, reaching the order of picoseconds. We focus on non-line-of-sight (NLOS) imaging, which attempts to reconstruct fully or partially occluded scenes by measuring the back-scattering of the hidden objects via

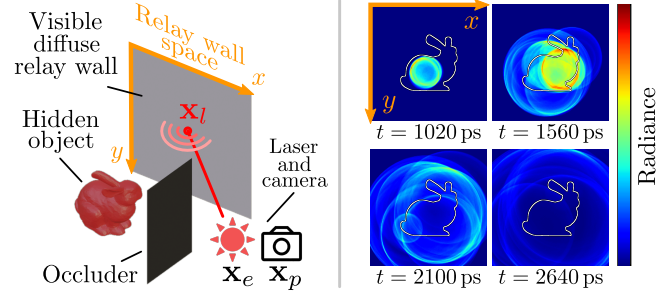


Figure 1: Left: A laser pulse is emitted towards a visible diffuse wall, scattering and illuminating the hidden object. Light reflects back to the wall and is captured by an ultra-fast time-resolved camera. Right: rendered simulation.

a visible relay wall (Figure 1). NLOS imaging has applications in autonomous driving, medical imaging or space exploration. These scenarios usually need expensive hardware, including pulsed illumination and ultra-fast cameras, difficult to calibrate and operate. Transient light transport simulation thus emerges as an alternative tool for developing, prototyping, and testing such systems.

We base our work in the transient path integral formulation by Jarabo et al. [2014]. In a NLOS setup, the geometry of interest and emitters are hidden behind multiple bounces, so light paths generated with a conventional algorithm are sub-optimal. We develop three sampling techniques for generating eye, light and hidden geometry subpaths with *virtual* irradiance meters and light sources. Our techniques can improve the convergence of NLOS transient path tracing by two orders of magnitude for equal render times.

Some of the existing NLOS capture simulation methods are limited to three-bounce paths [Chen et al. 2020; Iseringhausen and Hullin 2020; Tsai et al. 2019]. Rendering longer paths could allow to tackle more challenging problems, such as seeing behind two corners. We implement our methods in the Mitsuba 2 rendering system [Nimier-David et al. 2019], using our extension for transient rendering, which provides many advantages over previous transient path tracing systems [Jarabo and Arellano 2018; Jarabo et al. 2014; Pediredla et al. 2019], including support for CPU and GPU parallelization, simulation of the polarization of light, and the possibility of extending its differentiable rendering capabilities to the NLOS domain. Up to our knowledge our implementation is the first comprehensive and public¹ transient extension to Mitsuba 2.

¹<https://github.com/diegoroyo/mitsuba2-transient-nlos>

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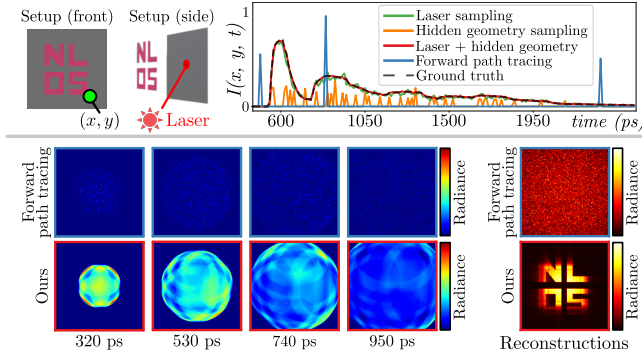


Figure 2: Top: Radiance over time for the (x, y) pixel in green, comparing techniques with equal samples. Ground truth uses two orders of magnitude more samples. Bottom: Radiance on the relay wall at different times t . Renders are processed for reconstructions using work by Liu et al. [2019].

2 OUR APPROACH

We develop three subpath sampling strategies, basing our work on the path contribution function for transient path tracing formulated by Jarabo et al. [2014]. A path $\bar{\mathbf{x}} = \mathbf{x}_0 \dots \mathbf{x}_k \in \Omega$ is defined as a vertex sequence from the light source $\mathbf{x}_0$ to the camera sensor $\mathbf{x}_k$. The light measured for a path, $f(\bar{\mathbf{x}})$, is proportional to:

$$f(\bar{\mathbf{x}}) \propto L_e(\mathbf{x}_0 \rightarrow \mathbf{x}_1, \Delta t_0) W_e(\mathbf{x}_{k-1} \rightarrow \mathbf{x}_k, \Delta t_k), \quad (1)$$

where L_e is the light emitted from $\mathbf{x}_0$ towards $\mathbf{x}_1$ at time Δt_0 , and the sensor importance W_e models spatial, angular, and temporal sensitivity in the same way. For additional details, we refer the reader to the original work [Jarabo et al. 2014]. Also, in a forward path tracer, the probability of each path, $p(\bar{\mathbf{x}})$, can be expressed as:

$$p(\bar{\mathbf{x}}) = p(\mathbf{x}_k) \prod_{i=0}^{k-1} p(\mathbf{x}_i | \mathbf{x}_{i+1}). \quad (2)$$

We look at the characteristics of sensors, light and hidden geometry and introduce three sampling techniques to generate vertices with distributions that fit NLOS scenes. We denote these techniques as *relay wall space*, *laser* and *hidden geometry* sampling, respectively. Figure 2 shows the effectiveness of our techniques.

2.1 Relay wall space

Consider a time-resolved image $I(x, y, t)$ where t denotes the temporal domain. The key difference between conventional line-of-sight (LOS) and NLOS images lies in the topology of (x, y) . While LOS images aim to reconstruct a continuous signal, NLOS captures use sparse points in the visible surfaces, with sensors that only measure light coming from a differential solid angle. We define a mapping from a parameterized 2D relay wall coordinates (x_s, y_s) to its corresponding 3D point as $\mathcal{M}_s(x_s, y_s) \rightarrow \mathbf{x}_s$. Thus, W_e becomes:

$$W_e(\mathbf{x}_{k-1} \rightarrow \mathbf{x}_k, \Delta t_k) = \delta(\mathbf{x}_p - \mathbf{x}_k) \delta(\omega_s - \omega_{k-1}) K_t(\Delta t_k), \quad (3)$$

with $\delta(\cdot)$ the Dirac delta, $\mathbf{x}_p$ the position of the sensor directed towards $\omega_s = \text{norm}(\mathcal{M}_s(x_s, y_s) - \mathbf{x}_p)$, a normalized direction, and $\omega_{k-1} = \text{norm}(\mathbf{x}_{k-1} - \mathbf{x}_k)$ and $K_t(\Delta t_k)$ the temporal importance. The distribution of these vertices for each path, i.e. $p(\mathbf{x}_{k-1} | \mathbf{x}_k)$, is uniform and regular along the relay wall parameterized 2D space.

2.2 Laser sampling

In a typical NLOS capture setup, light is emitted from a laser towards a point in the relay wall defined by the mapping $\mathcal{M}_l(x_l, y_l) \rightarrow \mathbf{x}_l$, resulting in an emission in a differential solid angle modeled as:

$$L_e(\mathbf{x}_0 \rightarrow \mathbf{x}_1, \Delta t_0) = \delta(\mathbf{x}_e - \mathbf{x}_0) \delta(\omega_l - \omega_0) E(\Delta t_0), \quad (4)$$

with $\mathbf{x}_e$ the position of the laser, $\omega_l = \text{norm}(\mathbf{x}_e - \mathcal{M}_l(x_l, y_l))$ and $\omega_0 = \text{norm}(\mathbf{x}_0 - \mathbf{x}_1)$, and $E(\Delta t_0)$ the emitted radiance over time. Paths in the search space Ω generated using forward path tracing have a zero probability of finding the light source. Using a tiny area light makes the result converge very slowly (Figure 2). Bidirectional path tracing could be used, however, our approach is simpler.

In *laser sampling*, the light contribution $\hat{L}(\mathbf{x}_0 \rightarrow \mathbf{x})$ to any point $\mathbf{x}$ through $\mathbf{x}_l$ can be analytically computed connecting the three vertices with probability $p(\mathbf{x}_1 | \mathbf{x}_2) = \delta(\mathbf{x}_l - \mathbf{x}_1)$. This method ignores longer paths (e.g. interreflections) and should be combined with classical sampling strategies in multiple importance sampling.

2.3 Hidden geometry sampling

The back-scattering of the hidden geometry can only be found through an indirect bounce on the relay wall. In a forward path tracer, generated vertices follow a distribution $p(\mathbf{x}_j | \mathbf{x}_{j+1})$ which targets the reflectance distribution or the light source location, both suboptimal in NLOS scenes. For each path, we randomly generate a vertex $\mathbf{x}$ on the surface of the hidden geometry with uniform probability, $p(\mathbf{x}_{k-2} | \mathbf{x}_{k-1}) = 1/A$, where A is its total surface area.

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